



Project Completion Report

(Dead-man-switch) Permission based Wallet, shipped as Epora

Project Name	(Dead-man-switch) Permission based Wallet, shipped as Epora
Project URL	projectcatalyst.io , Fund 11, Cardano Use Cases: Concept
Project Number	1100002
Project Manager	Sandro Schaier
Project Start Date	11 March 2024
Project Completion Date	26 September 2026
Plan	7 months, with deliveries from April to October 2024
Budget	100,000 ADA in five milestones
Completion video	Watch the close-out video on YouTube

Evidence labels. Verified means checked against the linked source or public chain data on 26 September 2026. Reported means supplied by the project manager or an earlier validation record. Planned work is not a delivered result.

1 Deliverables

Epora implements a permission-based smart wallet on Cardano. A normal wallet signs actions and pays transaction fees. The smart wallet holds funds and enforces its configured permissions. The public repository contains the Aiken contracts, web application, API, whitepaper, and tests. The project's own code uses the MIT license.

A. Output: Mainnet beta prototype

Delivered. The mainnet beta is online. Confirmed transactions show wallet creation, funding, an operator spend, and a settings update. The settings update added a beneficiary and a 30-day proof-of-life period. A later operator spend advanced that deadline. This is evidence of automatic renewal during an operator action, not the separate manual renewal path.

Acceptance criteria. A working mainnet prototype. The approved milestone specifies a beta because the code has not been externally audited.

Verified evidence. [Mainnet application](#); [health endpoint](#); [transaction hashes, balances, and measured fees](#). At 01:30 UTC, health returned HTTP 200 with database and indexer up. The reference-script endpoint returned ready. Section 2 gives the on-chain evidence.

B. Output: Bug fixes and categorized feedback

Delivered. The feedback register groups reports and links them to fixes or a deferral. Its 22 September snapshot contains 49 distinct issues: 46 categorized as fixed, one deferred, and two non-bugs. The fixed groups cover external reports, bugs, interface improvements, and operations. GitHub and Discord reports led to changes in the app.

Acceptance criteria. Aggregated, categorized feedback and bugs, plus further bug fixes and improvements.

Verified evidence. [Feedback register](#); [external report #561](#) and [#562](#); [Discord feedback #565](#). Counts describe the register, not a fresh retest of every fix. The VESPR retest is complete according to the project manager (*reported*).

C. Output: Communication about wallet integration

Delivered. We contacted wallet developers about a native integration and a possible smart-wallet CIP. The public Eternl Discord message dated 24 September introduces Epora and asks about integration interest. A repository screenshot makes the message available without Discord sign-in. Other contacts and an Eternl DM reply are recorded by the project manager (*reported*). Integration discussions continue.

Acceptance criteria. Communication with potential wallet developers, if developers or the community express interest. The approved requirement asks for communication about potential collaboration. It does not require a completed native wallet integration.

Verified evidence. [Outreach record](#); [public screenshot](#); [Discord message, sign-in required](#). The screenshot proves the sent message, not a partnership agreement.

D. Output: Project Completion Report

Delivered. This report describes the completed work, mainnet activity, feedback, and continuing maintenance. It follows the current Catalyst headings: Deliverables, Usage, Impact, and Sustainability. It links claims to evidence and keeps operational checks separate from measured results.

Acceptance criteria. A Project Closeout Report, accessible through a link in the final milestone submission.

Evidence. [This PDF](#); [source and evidence folder](#); [machine-readable mainnet snapshot](#).

E. Output: Project Completion Video

Delivered. The project manager supplied the YouTube video *Epora Wallet Catalyst-Closeout*. Review it alongside the live prototype and transaction evidence. The report and repository README link to the same video.

Acceptance criteria. A Project Closeout Video, linked in the report and final milestone submission.

Verified link. [Watch the project completion video](#). YouTube’s public metadata returned its title and the channel *schaier-io*. This checks availability, not a frame-by-frame review or Catalyst approval.

Earlier approved submissions

Verified on the Milestone Module on 26 September 2026. These submissions use lettered outputs, descriptions, acceptance criteria, and labeled evidence links. This report uses the same evidence pattern for Outputs A to E.

Milestone	Delivered scope and approved submission	Admin signoff
1	Social accounts, whitepaper, and development planning	7 July 2026
2	On-chain contracts, developer documentation, and security evidence	6 August 2026
3	UI walkthrough, application, public API, and code documentation	21 September 2026

Milestone 4 had zero PoA approvals at this check. Milestone 5 is the current submission. Earlier approvals do not constitute approval of this report.

2 Usage

Verified snapshot: 26 September 2026, 01:30 UTC. The current deployed policy identifies two active smart wallets. Queries covered each wallet payment credential, including possible stake-address variants. There were four unspent wallet outputs. The inventory excludes older policy versions and does not measure unique people or independent adoption.

Wallet	Wallet ADA	State deposit ADA	Total ADA
Smart wallet	10.000000	1.586080	11.586080
Smart wallet Demo	17.000000	2.202410	19.202410
Total	27.000000	3.788490	30.788490

Neither wallet payment balance contains other native tokens. Each separate state output holds its identifying state-thread token. The shared reference output holds another 63.270800 ADA. These deposits are separate from user wallet balances and are not fees or revenue.

Observed actions and measured fees

Verified through Koios: seven wallet transactions and one reference deployment have confirmed block records. Returned Plutus contract results are `valid_contract: true`. Plain funding transactions have no Plutus execution result. Fees below are ledger fees, not estimates.

Confirmed action	Fee ADA	Bytes	Transaction
Reference deployment	0.809617	14868	fb1186638076...
Wallet creation + funding	0.447150	838	71ea33e0e820...
Wallet additional funding	0.170825	350	d6e2e824daef...
Wallet operator spend	0.895889	10117	8d21e3b7f954...
Demo creation + funding	0.447194	839	6ba0055f3cdd...
Demo additional funding	0.170825	350	9e60d91d7100...
Demo beneficiary + liveness settings	0.478150	919	4080136da4b6...
Demo spend + liveness refresh	0.912502	10256	a9502540ad8e...

The Demo spend paid 10 ADA to its recipient and returned 10 ADA to the smart wallet. An earlier 7 ADA output remained unspent. The proof-of-life deadline moved from 25 October 2026, 23:27:06.972 UTC to 23:33:36 UTC. This verifies automatic refresh during an operator spend.

Current STT policy and validator hash:

0dac00be80879dcf585cdb9d0acf6e0ecf52c417ded30d8b72d0ebf1

Active reference transaction, output 0:

fb11866380769867300d1919098c5961473e362d40d03192f5b83c61eb6f0c0d

Reproduction and limits. The [evidence record](#) contains complete hashes, query methods, addresses, and limits. Cardanoscan blocked automated access. Its links point to the observed hashes; [Koios](#) supplied the verified data. These transactions do not establish a backup restore, an empty-database reindex, or every possible contract action.

3 Impact

The project gives Cardano users a contract-based way to configure shared control, spending permissions, and recovery after inactivity. A normal wallet keeps the signing keys. The smart wallet checks which action those keys may perform. A lost-key or inheritance plan can therefore include another authorized signer instead of relying only on one seed backup.

Challenge and project outcomes

The challenge brief sets aims rather than numeric adoption targets. The project measures delivery through its five milestones. The linked artifacts support the outcomes below.

Intended outcome	Delivered result and evidence
Proof of concept and design	Whitepaper v1.0 , executable Aiken validators, and reference applications on Preprod and mainnet.
Useful Cardano functions	Configurable operator permissions, weighted multisignature, allowances, proof of life, beneficiary recovery, and streaming payments. See the transition map and approved Milestone 2 submission.
Reusable specifications	Public API guide , OpenAPI specification , code documentation, and MIT license for project code.
Feedback improves the app	The register links 46 fixed issues to changes across connection handling, bugs, interface behavior, and operations. Two outside tester reports appear in the register.
Path to wallet integration	Public outreach and an API give wallet developers a starting point. A native integration or agreed partnership is not claimed.

Testing and engineering results

Reported validation record. The [release record](#) gives the exact output total: 750, passed: 750, failed: 0 for Aiken v1.1.23. It records 713 unit tests, 37 property tests, and a run with 10,000 cases per property. The blueprint was rebuilt and compared with the committed artifact. These are recorded results from 25 September, not a new test run for this report. The [security evidence map](#) connects stated threats to executable tests. None of this is an external audit.

Design result. One contract configuration combines the supported permissions. The state validator checks the declared action and state change. The wallet validator limits how funds move. This separates permission rules from payment enforcement and supports review through the published transition map.

Measured mainnet result. The prototype has completed real mainnet transactions. The measured Demo wallet creation fee was 0.447194 ADA. Its operator spend with automatic liveness refresh cost 0.912502 ADA. These measurements establish the observed costs only. They do not establish savings against other wallets or a fee bound for every configuration.

Timeline. The original plan covered seven months, ending in October 2024. Admin signoff for milestones 1 to 3 occurred in July through September 2026. The reported completion date is 26 September 2026. The approved project budget remains 100,000 ADA. The portal showed 59,000 ADA distributed at this check, so the report does not claim that the full budget has been received.

4 Sustainability

Ongoing project. Epora continues as an open-source project. Sandro Schaier and the project team plan to maintain the beta, triage public issues, and fix defects. The public repository and documentation let other developers inspect and contribute to the work.

Operating funds, reported by the project manager. Received Catalyst funds pay for servers and other operating costs. No fixed operating runway or committed recurring revenue is established in this report. Future funding or revenue depends on adoption and partner agreements.

Planned roadmap. Continue the beta with its permanent unaudited-code notice. Follow up with wallet teams and explore a smart-wallet CIP. The proposed standard could describe how wallets find smart wallets, display their rules and balances, and build permitted actions. Shared maintenance with wallet teams is a possible longer-term outcome, not an agreed commitment. External audit funding and scope remain future decisions.

Open checks and deferred work

Verified record status. The mainnet evidence closes the earlier gap around observed transactions and some actual fees. It does not establish the following operational checks:

- A timed backup restore, snapshot retention verification, or timed reindex from an empty database.
- Production database isolation and a deliberate monitoring-alert drill.
- Mainnet measurements for all remaining action types, multi-asset minimum ADA, and fragmented-wallet consolidation.

These remain in the [engineering checklist](#). The approved milestone acceptance criteria ask for a mainnet prototype, categorized feedback, wallet communication, the report, and the video. The checklist has broader operational work and does not itself amend those criteria.

Deferred, reported by the project manager. WalletConnect signing/device testing and wallet-signed terms acceptance remain deferred. The shipped beta acknowledgement is not wallet-signed terms acceptance. VESPR retesting, the external API guide review, and an API-built confirmed transaction are recorded as complete on the project manager's confirmation. The chain evidence here does not independently identify those tests or their client software.

Evidence and review links

- [Approved final milestone outputs, acceptance criteria, and evidence requirements.](#)
- [Public repository; mainnet evidence and limitations; feedback register.](#)
- [Wallet outreach; developer API guide; live API reference.](#)
- [Project Completion Video; plain-text A to E submission.](#)
- [Catalyst report and video guidance.](#)

Prepared on 26 September 2026 for Project 1100002. Sources: the Milestone Module, linked repository records, public deployment endpoints, and Koios chain data. The full query method and measurement limits accompany the report. Catalyst reviewers determine milestone acceptance.